

Credit Card Fraud Detection

End-to-end Machine Learning system tracking extreme class imbalance in real-world financial transaction data.

Python

Scikit-Learn

XGBoost

Streamlit

Fraud Detection Project | Kaggle ULB Dataset | Jihane Bachiri

Why *fraud detection* is a hard machine learning problem?

THE CORE CHALLENGE

Out of **284 807 transactions**, only **492 are fraudulent**, a ratio of 0.17%. A naive classifier that labels everything as legitimate would reach 99.83% accuracy while catching zero frauds.

WHY ACCURACY IS INVALID?

The dataset is **579× imbalanced**. Accuracy rewards ignoring the minority class entirely.

BUSINESS FRAMING

False negatives (missed fraud) carry higher cost than false positives (blocked legitimate transactions). Threshold tuning lets us control this trade-off explicitly.

METRICS RETAINED

1. **Recall** - Minimize missed frauds. Each false negative is a real financial loss for the customer.
2. **F1-score** - Harmonic mean of precision & recall. Balance catching fraud without flooding the ops team with false alerts.
3. **AUC-ROC** - Threshold-agnostic ranking quality. Reveals true discriminative power across all operating points.

Key findings from *exploratory analysis*

284k

Total transactions

492

Confirmed frauds

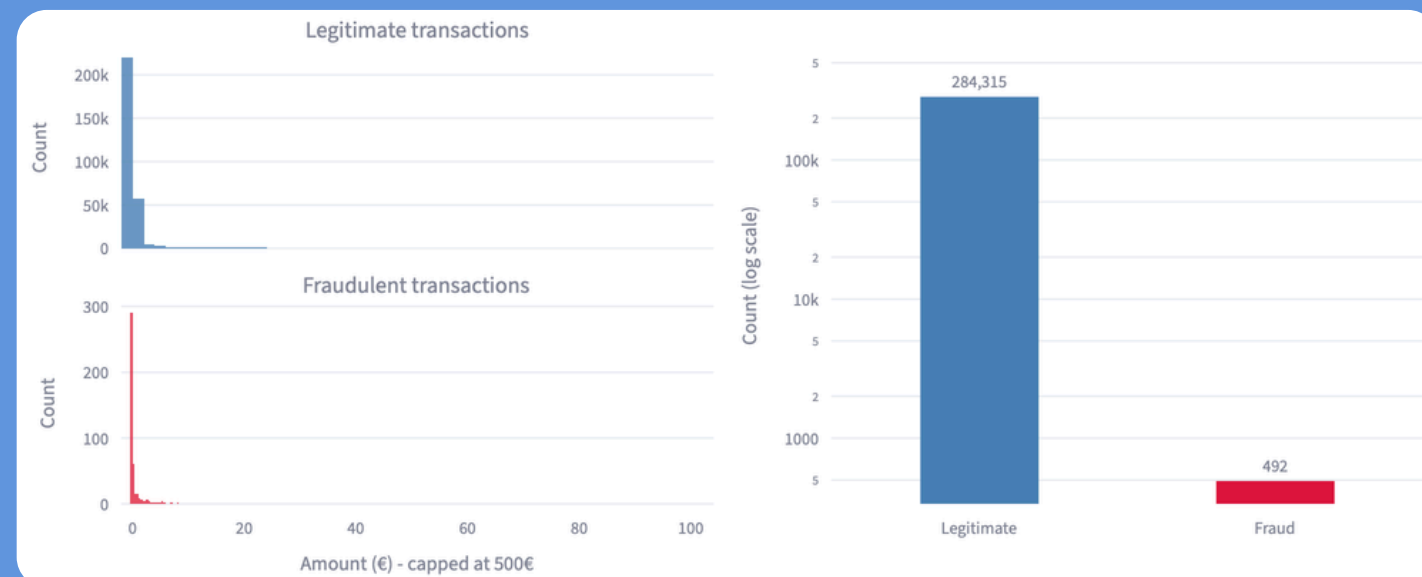
0.17%

Fraud rate

0

Missing values

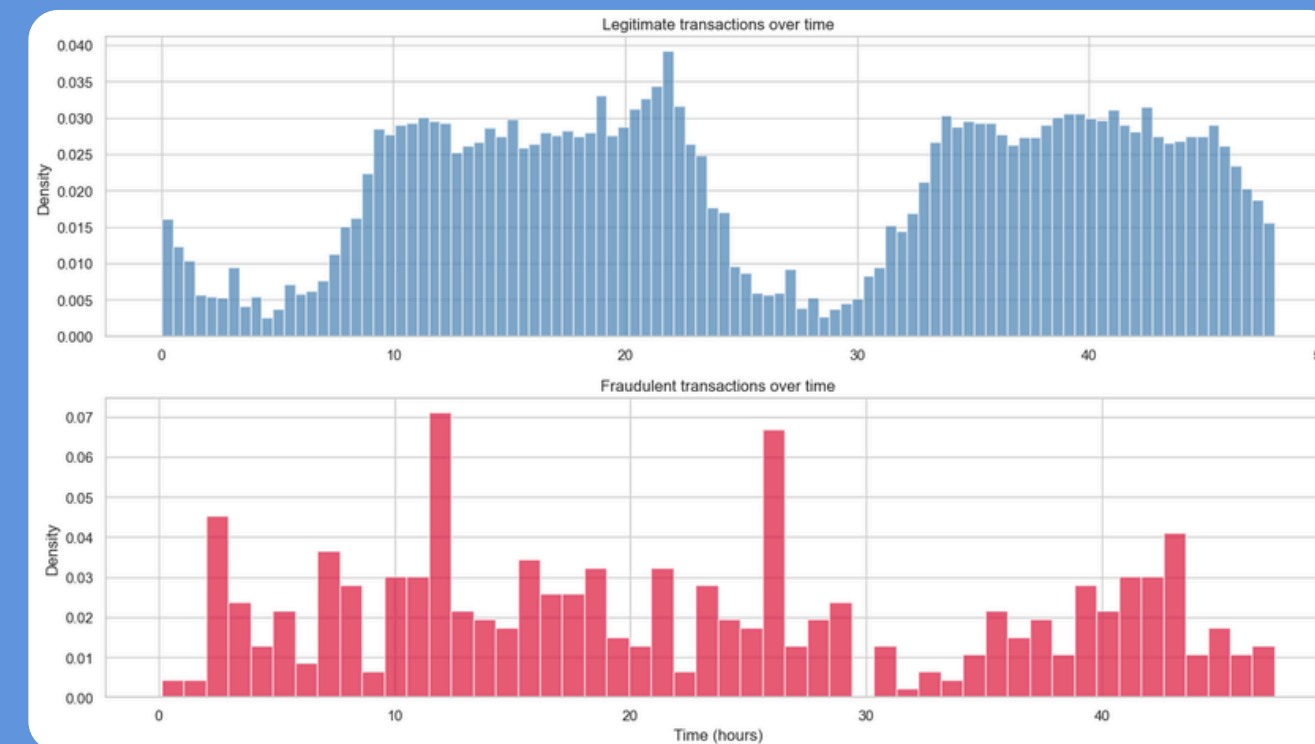
TRANSACTION AMOUNT DISTRIBUTION



Histogram - Amount (capped 500€)
Legitimate vs. Fraud (two subplots)

Frauds are **concentrated between 0 and 50€**. Median legitimate transaction: 22€, mean: 88€ (heavy right skew).

TIME DISTRIBUTION



Legitimate transactions follow a clear daily cycle. Frauds are more **uniformly distributed**, with a slight nighttime uptick.

MOST DISCRIMINATIVE FEATURES (CORRELATION WITH CLASS) V17, V14, V12, V10 and V11.

End-to-end pipeline & Key design decisions

01

EDA

Imbalance analysis
Correlation heatmap
Feature distributions

02

PREPROCESSING

StandardScaler
Stratified split 80/20
SMOTE on train only

03

SUPERVISED

LR · DT · RF · XGBoost
5-fold cross-val
Threshold tuning

04

UNSUPERVISED

PCA 2D
K-Means (k=5)
DBSCAN (outliers)

05

DASHBOARD

Streamlit + Plotly
Streamlit Cloud
Live deployment

SCALING

StandardScaler applied to *Amount*
and *Time* only.

V1-V28 are already PCA-normalized
by the dataset provider.

HANDLING IMBALANCE

SMOTE applied after train/test split
(exclusively on training data) to avoid
data leakage. XGBoost uses native
`scale_pos_weight = 577` instead.

CROSS-VALIDATION

No separate validation set.
Stratified 5-fold CV preserves the
original fraud ratio in each fold, giving
robust generalization estimates.



Model Comparison across *all metrics*

Model	Precision	Recall	F1-score	AUC-ROC	False Negatives	False Positives
Logistic Regression	0.06	0.92	0.11	0.96698	8	1458
Decision Tree	0.03	0.90	0.06	0.9522	10	2896
Random Forest	0.82	0.82	0.82	0.9688	18	17
XGBoost	0.88	0.85	0.86	0.9652	15	11

WHY XGBOOST WINS?

Best **F1-score** (0.86) and **fewest false positives** (11 vs. 17 for RF).

In production, false positives mean legitimate transactions blocked and frustrated customers: a direct operational cost.

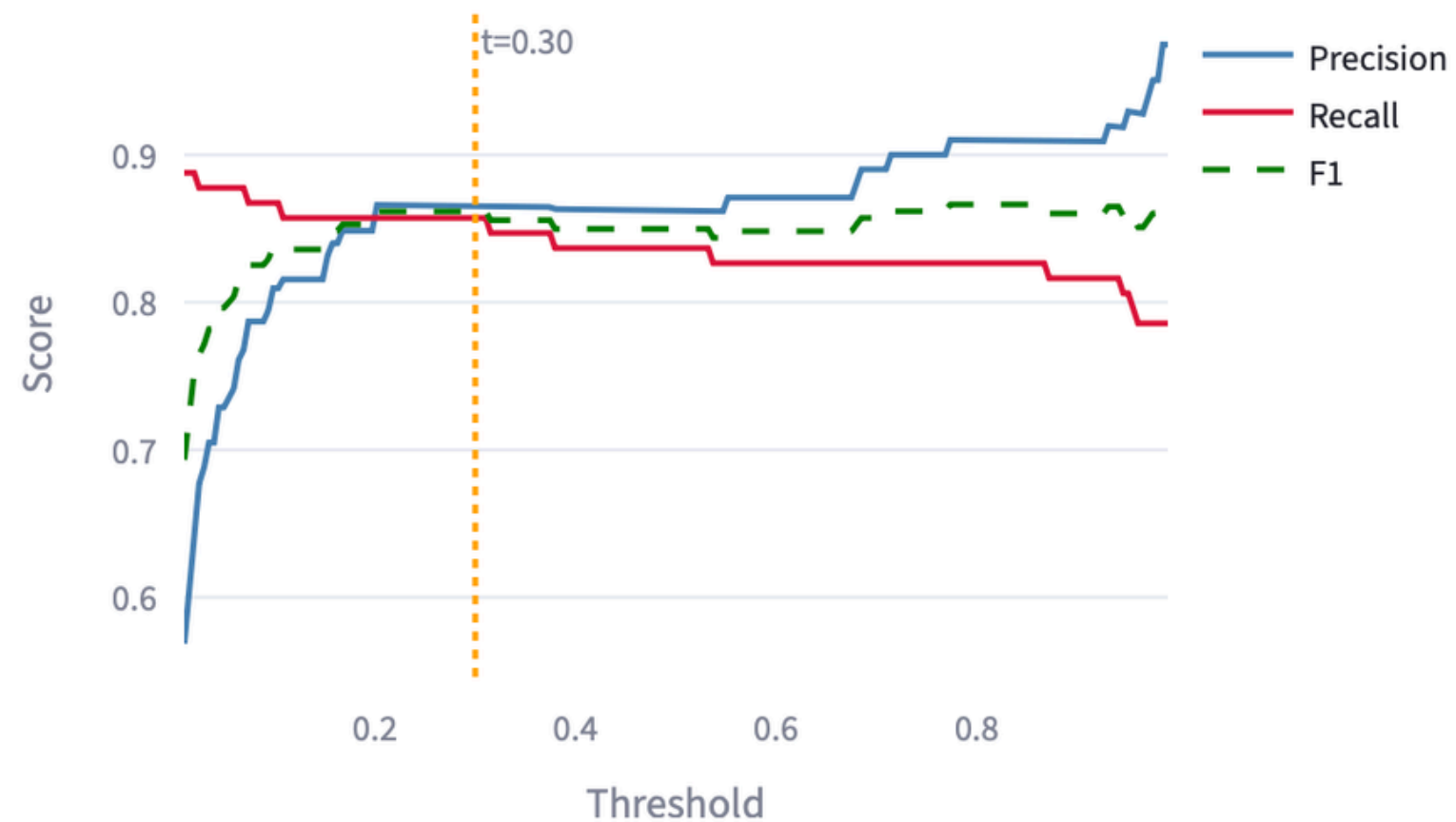
LOGISTIC REGRESSION / DECISION TREE TRADEOFF

High recall (≥ 0.90) but catastrophically low precision, generating **1 458–2 896 false alerts**.

SMOTE amplified the fraud signal too aggressively for simple models. XGBoost's native weighting mechanism is more controlled.

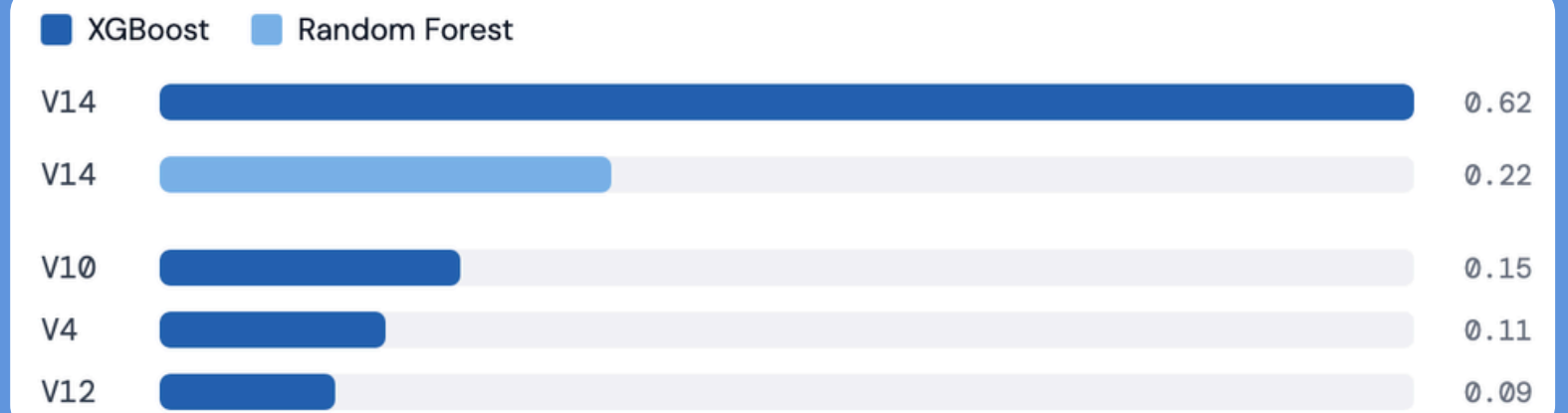
Threshold tuning & *feature importance*

PRECISION / RECALL VS. THRESHOLD



Precision · Recall · F1 curves
Vertical line at threshold = 0.30

FEATURE IMPORTANCE - XGBOOST VS. RANDOM FOREST



V14 dominates both models: 0.62 importance in XGBoost, 0.22 in RF. Strong consensus signal despite PCA anonymization. V10, V4, V12 follow consistently.

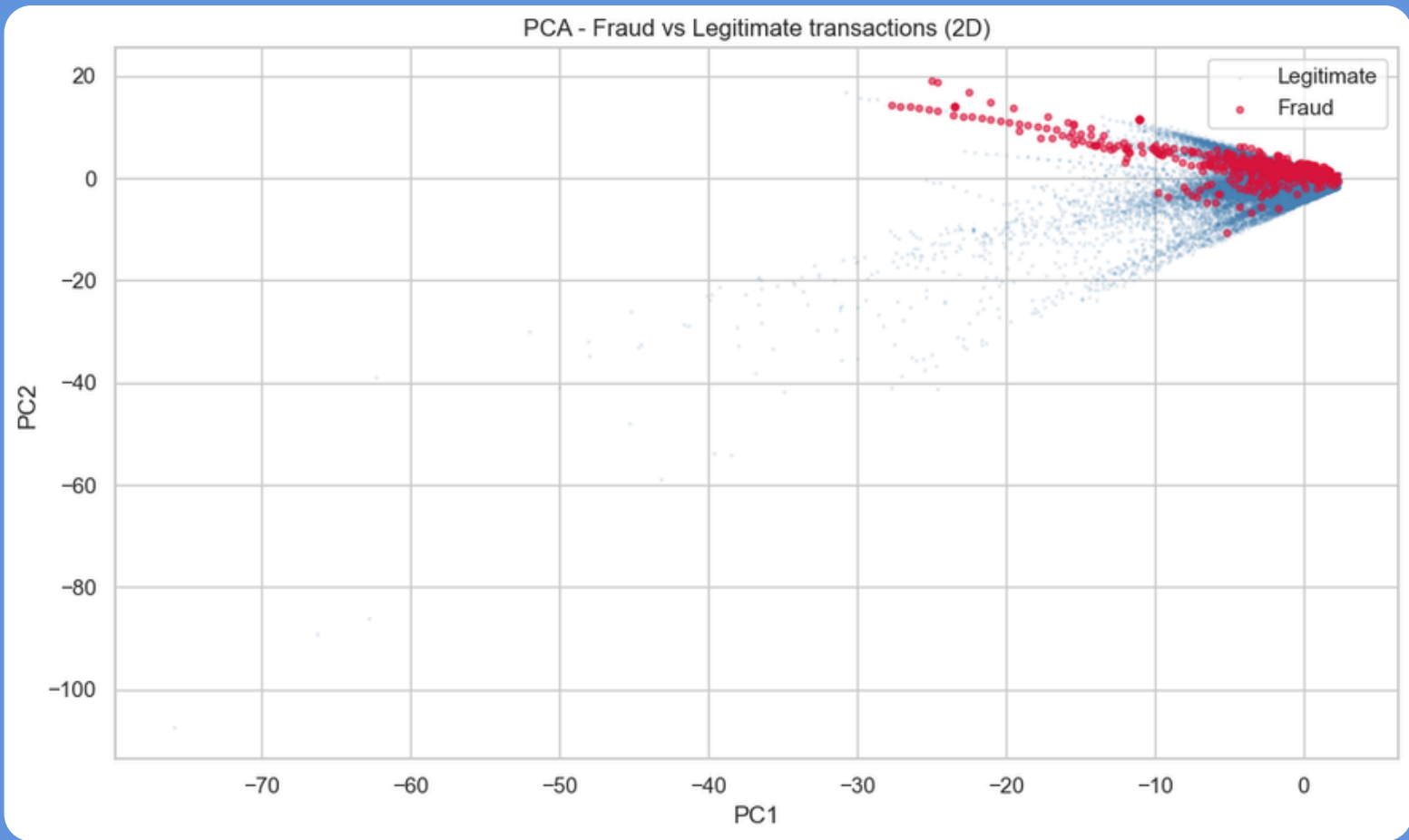
PRECISION / RECALL VS. THRESHOLD

Default threshold (0.5) misses too many frauds. Lowering to **0.30** improves recall at an acceptable precision cost, aligned with the business priority of catching fraud first.

Precision=0.88 ; Recall=0.85; F1-score=0.86

Clustering as a complementary fraud lens

PCA - 2D PROJECTION



Frauds cluster visually between **PC1 = -30 and 5**, confirming separability even in 2D despite losing 78% of information.

K-MEANS (K=5) - FRAUD CONCENTRATION BY CLUSTER

CLUSTER	TRANSACTIONS	FRAUDS	FRAUD RATE
C0	124 211	143	0.12%
C1	42 824	4	0.01%
C2	5 135	19	0.37%
C3 *	6 013	202	3.36%
C4	106 624	124	0.12%

Cluster 3 concentrates **41% of all frauds** in just 2.1% of transactions, a fraud rate **20× the global average**.

High-value candidate for rule-based pre-screening.

DBSCAN - OUTLIER DETECTION

220 outliers detected with **39.5% precision**. Run on 10k sample ($O(n^2)$ constraint). Modest performance expected, as 2D PCA loses too much signal. Unsurprising without hyperparameter tuning.

Interactive system live on *Streamlit Cloud*

DASHBOARD PAGES (5)

1

Business Overview

KPI cards, amount distribution, class balance

2

Model Comparison

Side-by-side metrics table + superposed ROC curves

3

Threshold Explorer

Interactive slider, real-time F1/FP/FN, business impact €

4

Feature Importance

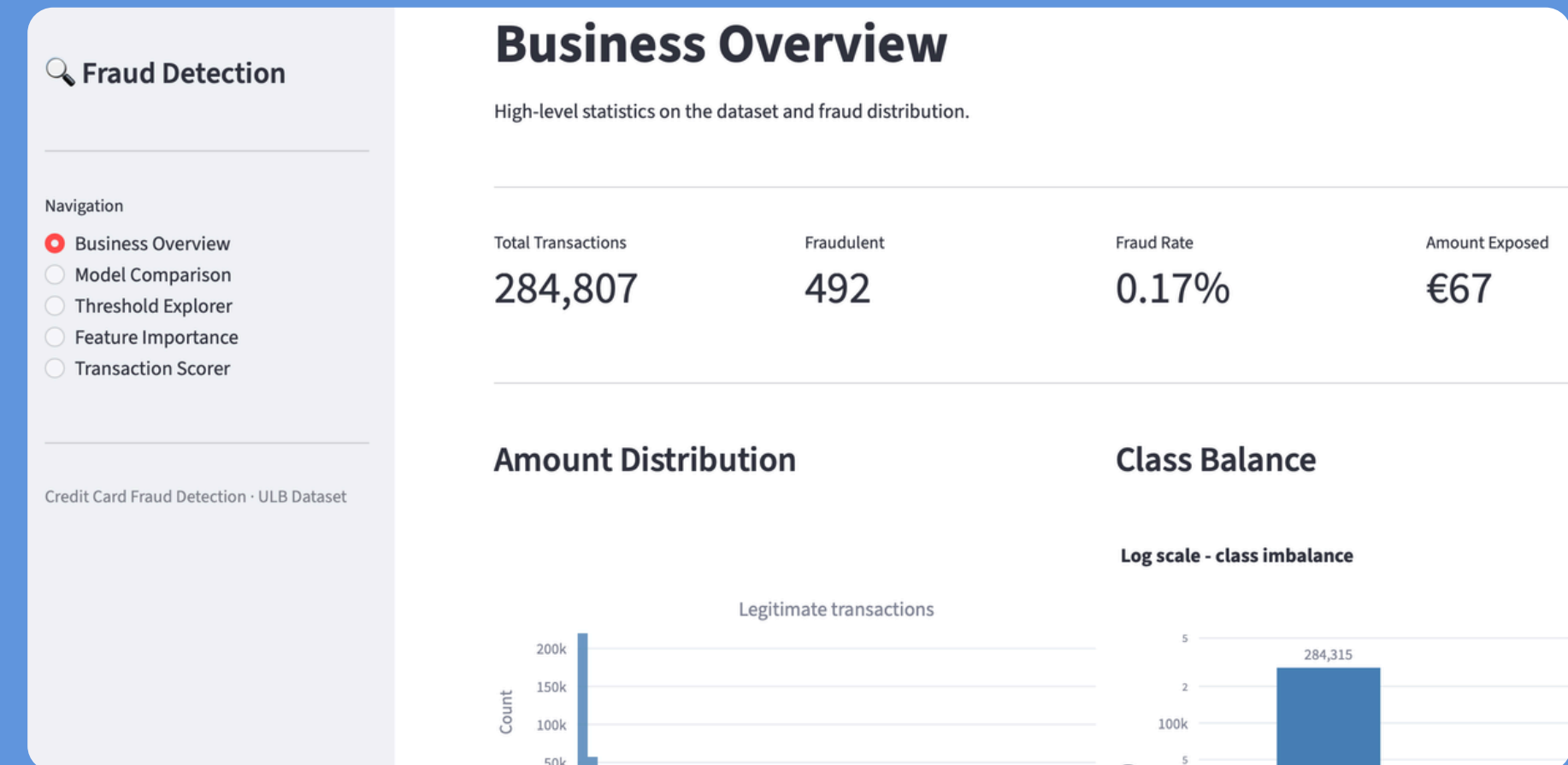
XGBoost vs. RF, consensus top features

5

Transaction Scorer

Manual input or demo fraud/legitimate, live gauge

SCREENSHOT



Click on the screenshot to access the live dashboard!

